DATA ITEM DESCRIPTION

Title: NUCLEAR SURVIVABILITY PROGRAM PLAN

Number: DI-NUOR-80156B

Approval Date: 20130201

AMSC Number: F9318

Limitation:

DTIC Applicable: N/A

GIDEP Applicable:

Preparing Activity: 27 (AFNWC)

Applicable Forms: N/A

Use/Relationship: The Nuclear Survivability Program Plan describes how the contractor's Nuclear Survivability Program shall be conducted. It describes operations, procedures, design, analyses, tests and management activities to be performed to meet the specific nuclear survivability requirements.

a. This Data Item Description (DID) contains the content, format and intended use information for the data product generated by the specific and discrete work task described in the contract.

b. This DID provides the basis for the Government's determination that the contractor's Nuclear Survivability Program will meet contractual requirements in a cost-effective manner and is applicable to developmental and non-developmental programs.

c. Application. When a solicitation or contract contains a requirement for nuclear survivability this DID shall be listed on the Contract Data Requirements List (CDRL) (DD Form 1423). The documents sited in this DID are to be used for guidance and are not mandatory for the CDRLs.

d. Air Force Instruction (AFI) 10-2607, *Air Force Chemical, Biological, Radiological and Nuclear Survivability*, contains the AF policy applicable to nuclear survivability.

(Copies of the AF Instruction are available online at http://www.e-publishing.af.mil/.)

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the solicitation or contract.

2. General. The Nuclear Survivability Program Plan shall specify the methods and techniques for incorporating nuclear survivability into design, development and equipment integration, and for conducting a comprehensive hardness assurance program. It shall include a discussion of the procedures and methods to implement hardness maintenance and surveillance of the deployed equipment. The program plan shall be prepared consistent with the information contained in AFI 10-2607, and shall include the information as described below.

3. Format. The Nuclear Survivability Program Plan shall be in the contractor's format.

4. Content.

4.1 Specifics described in the plan. The plan shall describe in detail the specific approaches necessary to achieve nuclear survivability. This discussion shall include electrical and mechanical hardening design features and any operational or procedural methods planned.

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a. The plan shall detail the combination of analyses and tests required to demonstrate survivability. A discussion of design margins to be employed in the analysis shall be included.

b. The program plan shall describe the nuclear survivability test program to be performed under contract. Testing at each level of integration (piece part, circuit, etc.) shall be discussed in terms of need, simulators, test levels, dosimetry, instrumentation, etc.

c. The plan shall include a discussion of the management of the survivability program. The duties and responsibility of primary point of contact shall be identified and discussed. The relationship between the survivability manager and the program manager shall be detailed.

d. The program plan shall include a functional and operational description of the end item being developed or acquired under the respective contract. This description shall address the key subsystem elements in sufficient detail to enable the government program manager to understand the nuclear survivability program approach.

e. The plan shall present a schedule of critical survivability program milestones related to the overall program schedule. This time-phased schedule shall indicate initiation, review and completion for each key nuclear survivability task.

f. The plan shall present a general discussion of the integrated logistic support management approach to field a survivable system, including design, analysis, test and documentation.

g. The plan shall provide the estimated labor hours, the caliber of labor and the costs for material, supplies, travel, computer and simulation facilities.

4.2 Designs, analyses, tests, and evaluations described in the plan.

a. The plan shall delineate analytical efforts that supplement or replace testing, to include identification of computer codes and the intended application of these analytical techniques.

b. The plan shall delineate piece part, component and subsystem tests to be performed for each specified nuclear environment with details, such as sample size, test methods, test instrumentation, parameters to be characterized and relationships to analytical efforts.

c. The plan shall delineate system level tests to be performed, including methods of extrapolation from test environments to the threat environments, the rationale for simulator choices and relationships to analytical efforts.

d. The plan shall describe anticipated test simulators and test equipment to be employed, test configurations, test article orientation, exposure levels and test data to be obtained.

e. The plan shall describe the survivability philosophy to include the basis for selection of piece parts, materials, device technologies, circuit and mechanical designs; the specific materials, technologies and designs rejected because of nuclear survivability considerations; assumptions concerning the system design margins; planned operational fixes such as cycling power and any assumptions concerning the system operation, function, deployment or configuration that has been used in developing the Nuclear Survivability Plan. There shall be particular emphasis on design philosophy for hardness critical items (HCI) and mission critical equipments. The plan shall describe the configuration control approach to control HCIs and harness critical processes (HCPs).

f. The plan shall describe the approach to the incorporation of nuclear survivability issues and consideration into the appropriate integrated logistic support documentation as early as possible in the program. The integrated logistic support implications shall include operational procedures, maintenance, repair and overhaul procedures, personnel skill levels, spare parts, repair tools, provisioning and training needs.

g. The plan shall delineate the projected requirements, driven by the nuclear survivability of the system, for custom or hardened parts, materials, or components; the basis of need for these

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parts; and the impacts of these requirements on the program and system design, cost, operation, function and deployment.

h. If the prime contractor or any of the subcontractors are supplying non-developmental or proprietary materials, components or equipments under this contract, this plan must detail how the nuclear survivability of these materials, components or equipments shall be evaluated, assured, maintained and documented during all acquisition phases.

i. The plan shall identify the nuclear survivability trade-offs that shall be performed against other system requirements such as schedule, cost, mobility, reliability, producibility and Integrated Logistic Support (ILS).

4.3 Risk analysis and identification described in the plan.

a. Nuclear survivability areas of high risk and uncertainty, including the impact on cost, performance, weight, and Manpower and Personnel Integration (MANPRINT) shall be identified.

b. Assumptions, conclusions, and reasons used in risk analysis and identification shall be delineated.

c. Actions to be taken by the contractor to minimize the impact of risk and uncertainties identified shall be delineated.

d. When failure modes and effect analysis (FMEA) or failure modes and effects criticality analysis (FMECA) is required elsewhere in the contract the plan shall indicate how the impact of the survivability degradation risks will be integrated.

4.4 Management interface controls described in the plan.

a. Interfaces between the Nuclear Survivability Program and other related programs, such as engineering design, reliability, maintainability, logistic support, configuration management, production engineering and producibility programs shall be described.

b. Methodology and procedures by which the prime contractor shall ensure that any appropriate nuclear survivability program is executed in subcontracts shall be described.

c. Actions to be taken to ensure the nuclear survivability (to include validation, assurance and maintenance) for materiel to be procured elsewhere, including any non-developmental or proprietary items, shall be delineated.

d. Special requirements to be included in source or product selection shall be delineated.

e. The management strategy for mission essential items which do not meet system or materiel survivability requirements shall be described. This should also be reflected in risk identification (4.3).

4.5 Program management described in the plan.

a. Responsibilities and authorities of the Nuclear Survivability Program Manager and the nuclear weapons effects experts assigned to the program shall be described.

b. Management procedures and controls to assure the successful application and accomplishment of nuclear survivability requirements shall be described.

5. Additional information. The plan shall provide any additional information necessary to adequately delineate the Nuclear Survivability Program.

6. End of DI-NUOR-80156B.

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